

TC100 PID Temperature Controller

SKU / Model Number: TC100-48-SSR
Manufacturer: ThermoControl Ltd
Category: Controllers | ETIM Code: EC001192

PRODUCT OVERVIEW

The TC100 is a 1/16 DIN PID temperature controller with auto-tuning for industrial heating and cooling processes. Features dual display, SSR/relay outputs, and RS485 communication.

TECHNICAL SPECIFICATIONS

Control Mode	PID with auto-tune
Input Types	TC (J,K,R,S,T,E,N,B), RTD (Pt100)
Accuracy	+/- 0.3% FS +/- 1 digit
Sampling Rate	100 ms
Display	Dual 4-digit 7-segment LED
Control Output 1	SSR drive (12 V DC, 20 mA)
Alarm Outputs	2x relay (3 A, 250 V AC)

ELECTRICAL RATINGS

Supply Voltage	100-240 V AC, 50/60 Hz
Power Consumption	5 W
Communication	RS485 Modbus RTU
Retransmission	4-20 mA (optional)
Isolation	1500 V AC (input to output)

DIMENSIONS & MECHANICAL

Panel Cutout	45 x 45 mm (1/16 DIN)
Bezel Size	48 x 48 mm
Depth behind panel	110 mm
Mounting	Panel mount with brackets
Weight	150 g

OPERATING CONDITIONS

Operating Temperature	0 to 50 deg C
Storage Temperature	-20 to 65 deg C
Humidity	20 to 90% RH
Vibration	IEC 60068-2-6
IP Rating	IP65 (front face)

MATERIALS OF CONSTRUCTION

Front Panel	ABS, UL94 V-0
Housing	PC, UL94 V-0
Display	Red/Green LED
Terminals	Screw terminals
Keypad	Silicone rubber

PERFORMANCE DATA / SELECTION TABLE

Input	Range	Resolution	Accuracy
TC Type K	-200 to 1370 C	0.1 / 1 C	+/- 0.3% +1
TC Type J	-200 to 1200 C	0.1 / 1 C	+/- 0.3% +1
Pt100	-200 to 850 C	0.1 C	+/- 0.2% +1
0-10 V	Scalable	0.1 / 1	+/- 0.3% +1
4-20 mA	Scalable	0.1 / 1	+/- 0.3% +1

CERTIFICATIONS & APPROVALS

CE | UL | CSA | RoHS | REACH

Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.

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